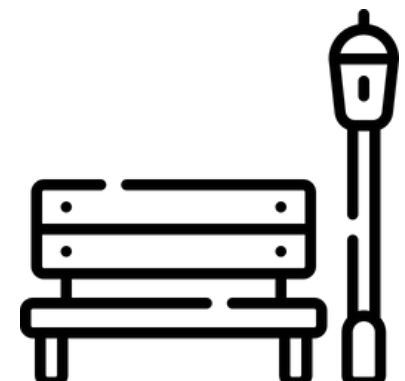


TEAM 3

SMART WASTE MANAGEMENT SYSTEM



Overflowing Urban Waste Bins



Overflowing Urban Waste Bins



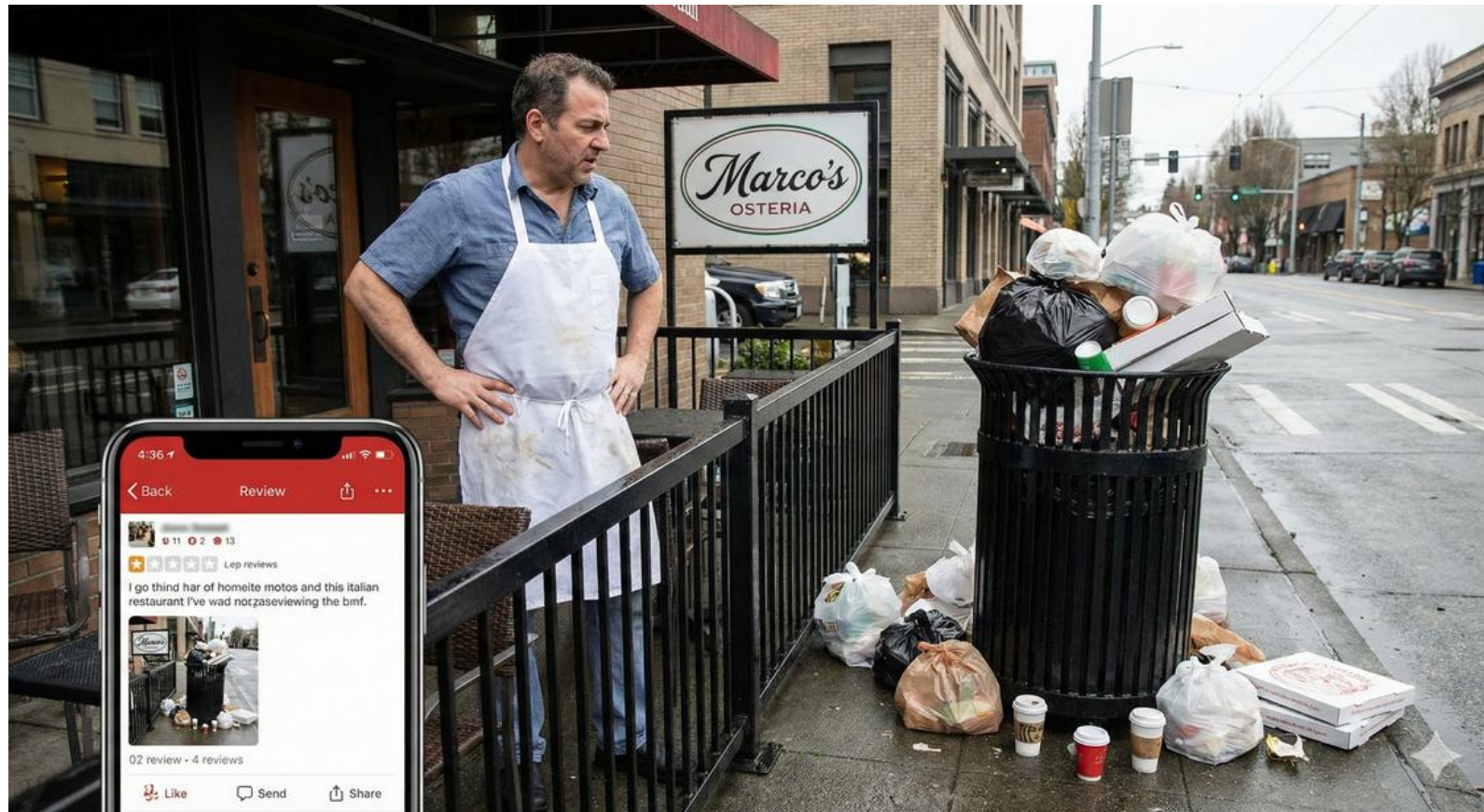
Urban Hygiene &
Visual Degradation



Inefficient Waste
Collection



Lack of Operational
Visibility



Name : Marco (53 years old)

Occupation : The owner-chef of an Italian restaurant operating an outdoor terrace in the city center

Responsibilities

- **Family livelihood:** This restaurant is the sole source of income for a family of five.
- **Food safety and compliance:** We have a legal and moral responsibility to pass rigorous health department inspections and provide customers with safe food.
- **Brand image management:** We take great pride in selling a 'luxurious and clean outdoor dining experience'.

Pain Points

- **“Waste beyond my control”:** A public trash bin installed by the city in front of the restaurant often overflows during busy weekends, leaving odors and visual discomfort that directly affect the business.
- **War with pests:** Overflow attracts rats and flies; one incident scared a customer away and led to negative Yelp and Google reviews, causing reservations to drop.
- **Sanitation inspection risk:** Although the trash is not from the restaurant, health authorities issued a warning that could lead to suspension, while the city’s response was slow and procedural.

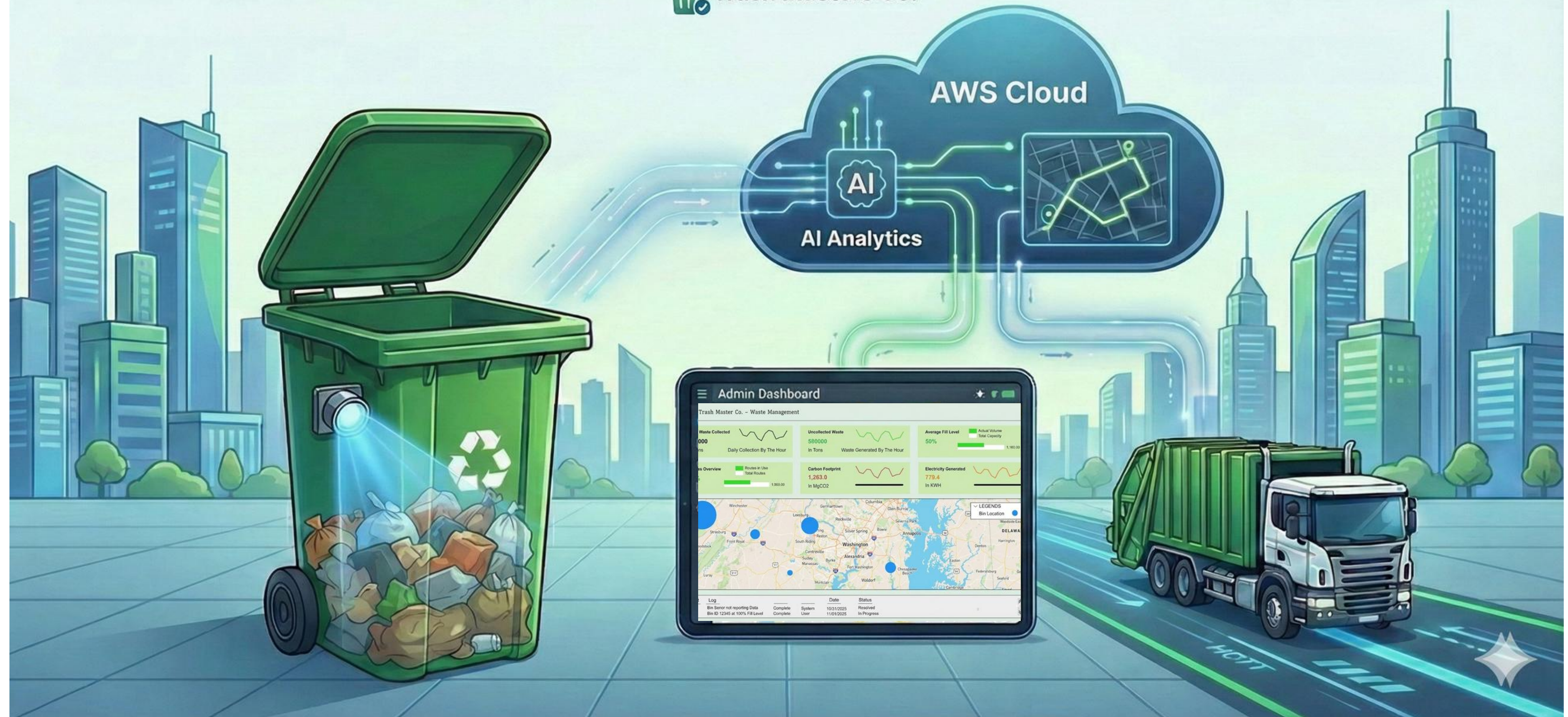
Needs

- **Predictive Alert System:** When a trash bin reaches 80% capacity, an alert is sent to the collection company to ensure the truck makes a priority visit to the storefront.
- **Clarification of Responsibility:** Hourly loading data is needed to show that an overflow is caused by collection delays, not by the business.
- **Impact of Citizen Reports:** Citizen reports (human sensors) should be treated not as simple complaints, but as urgent signals to prevent economic loss in commercial areas.

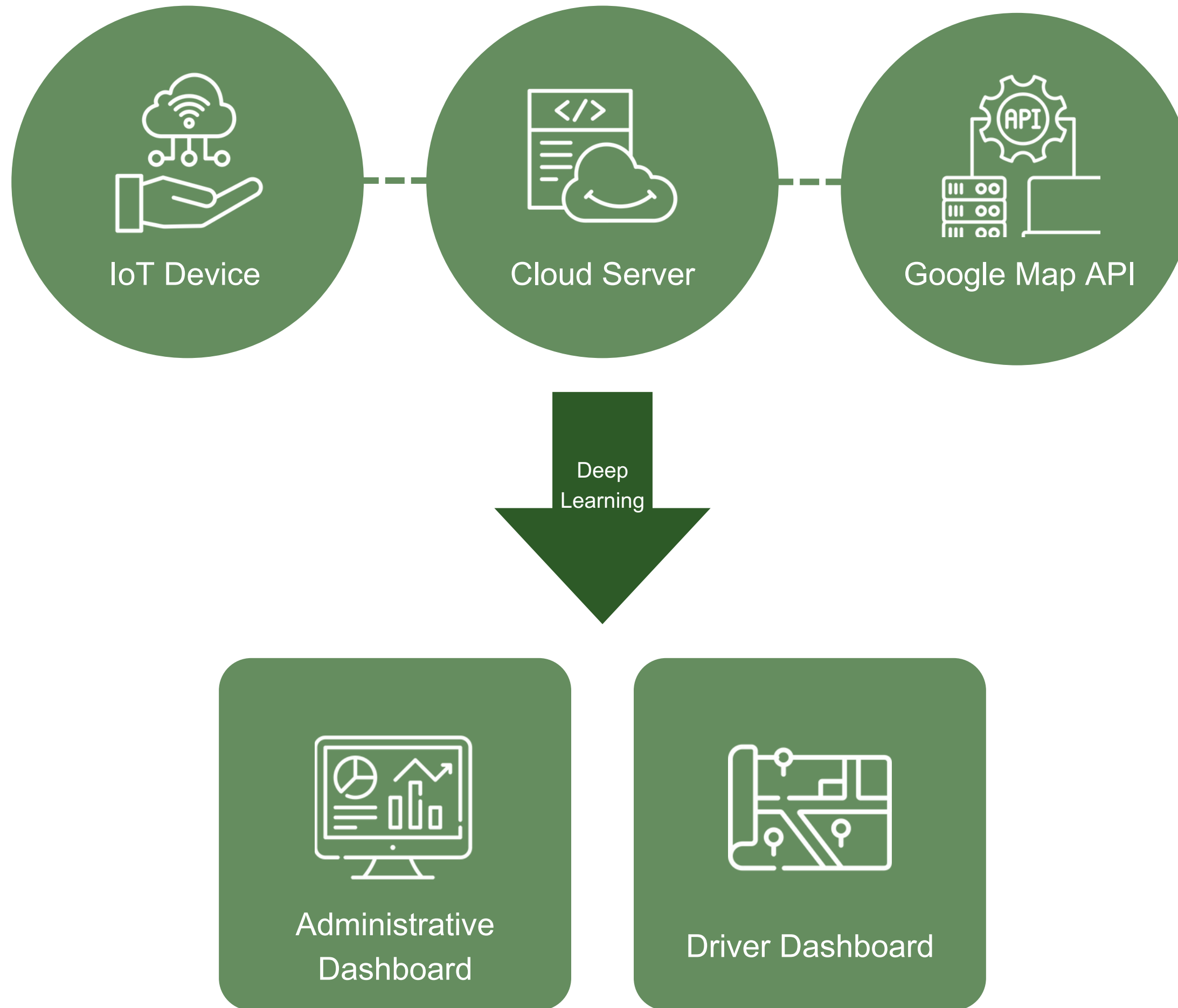
Smart Waste Management System



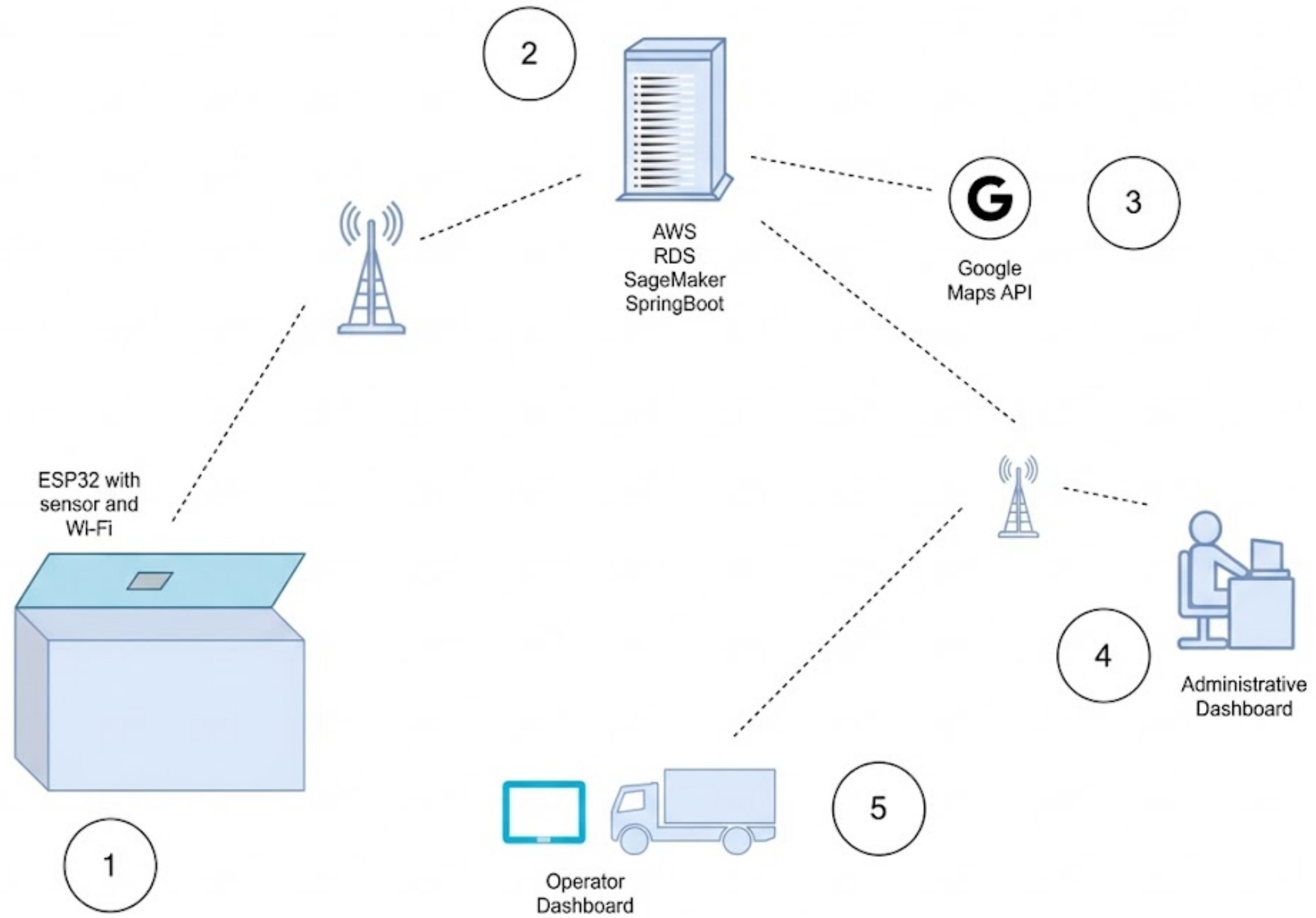
Trash Masters Co.



part 2
Solution



part 2
Solution



Economic Efficiency



Reduce empty pickups
Optimize fuel and labor
costs

Proactive Urban Management



Prevent bin overflows
Improve hygiene and public
satisfaction

Digital Transformation



Data-driven decision-making
Scalable waste management
system

1

People living in Seattle must be feeling inconvenienced by the overflow of trash.

2

Using our product will improve waste collection efficiency.

3

If our product can improve efficiency, city governments and waste collection companies will use it.

1

People living in Seattle must be feeling inconvenienced by the overflow of trash.

Citizen Survey

Location

Location: Seattle public spaces

Duration: 1–2 days

Method

Two short one-sentence questions
(≤30 seconds)

Target: Citizens

Question:

“Do you feel that overflowing trash bins
are a significant problem in this area?”

“What was the biggest issue?”

Record: Express opinions by placing
stickers on the panel

Validation Criteria

The hypothesis is supported if:
Majority of respondents express
inconvenience or concern

Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree

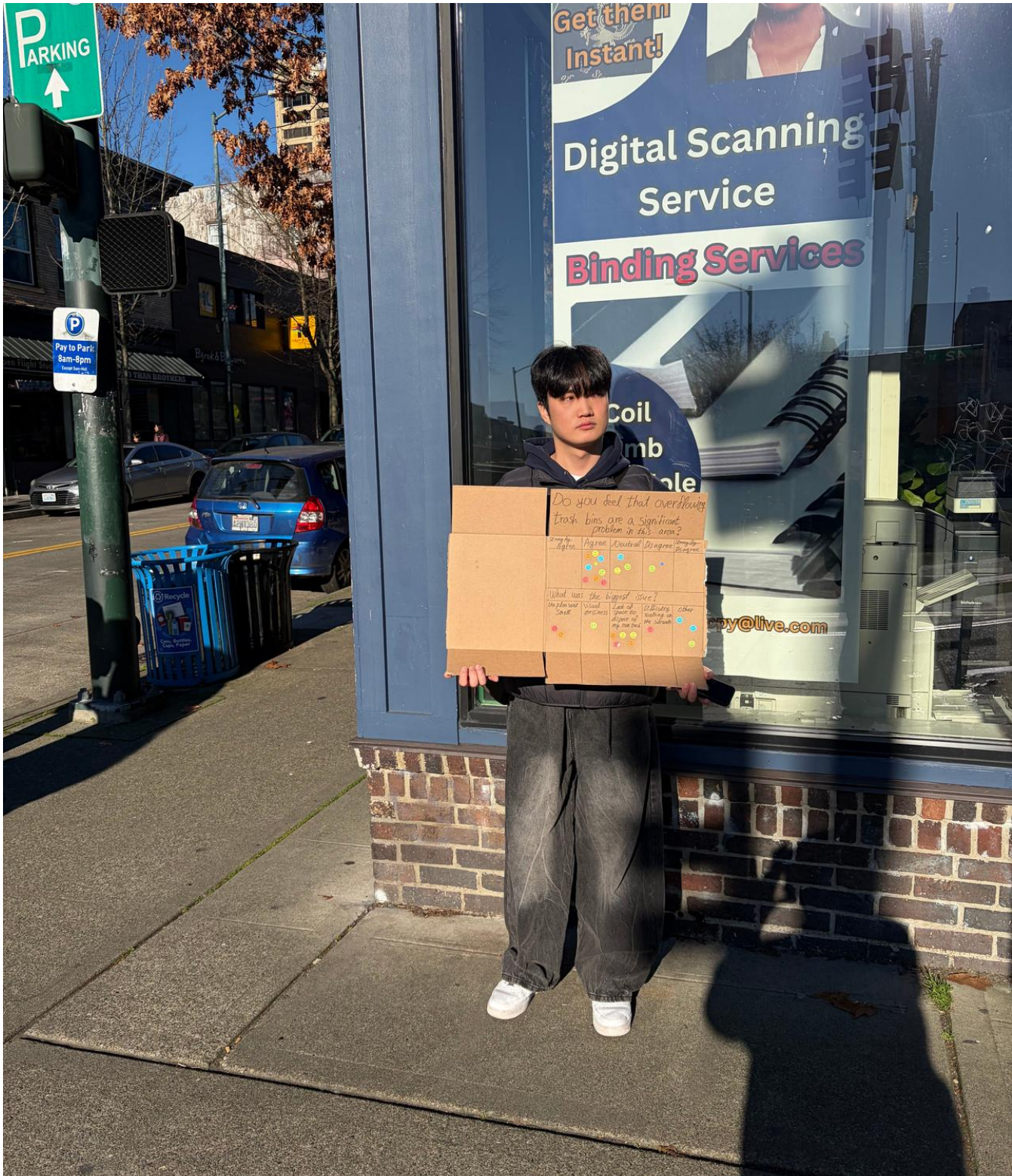
Unpleasant Smell 	Visual messiness 	Lack of space to dispose of my own trash 	Difficulty walking on the sidewalk 	Other 😊 
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Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
 	    	 		

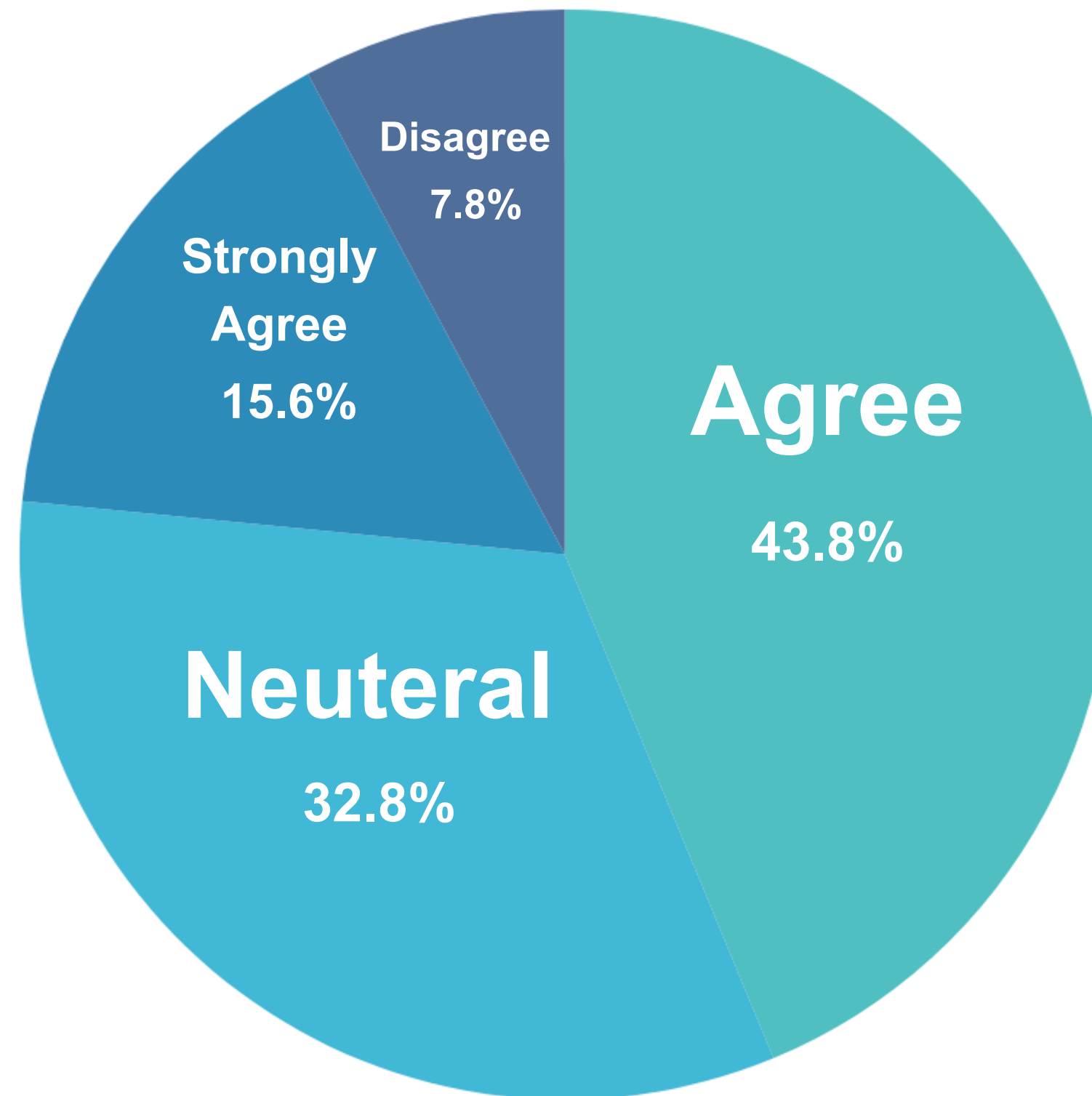
Inpleasant	Visual	Lack of	Difficulty	other
Smell	messiness	space to dispose of my own trash	Walking on the sidewalk	
 	   	  	 	

part 4

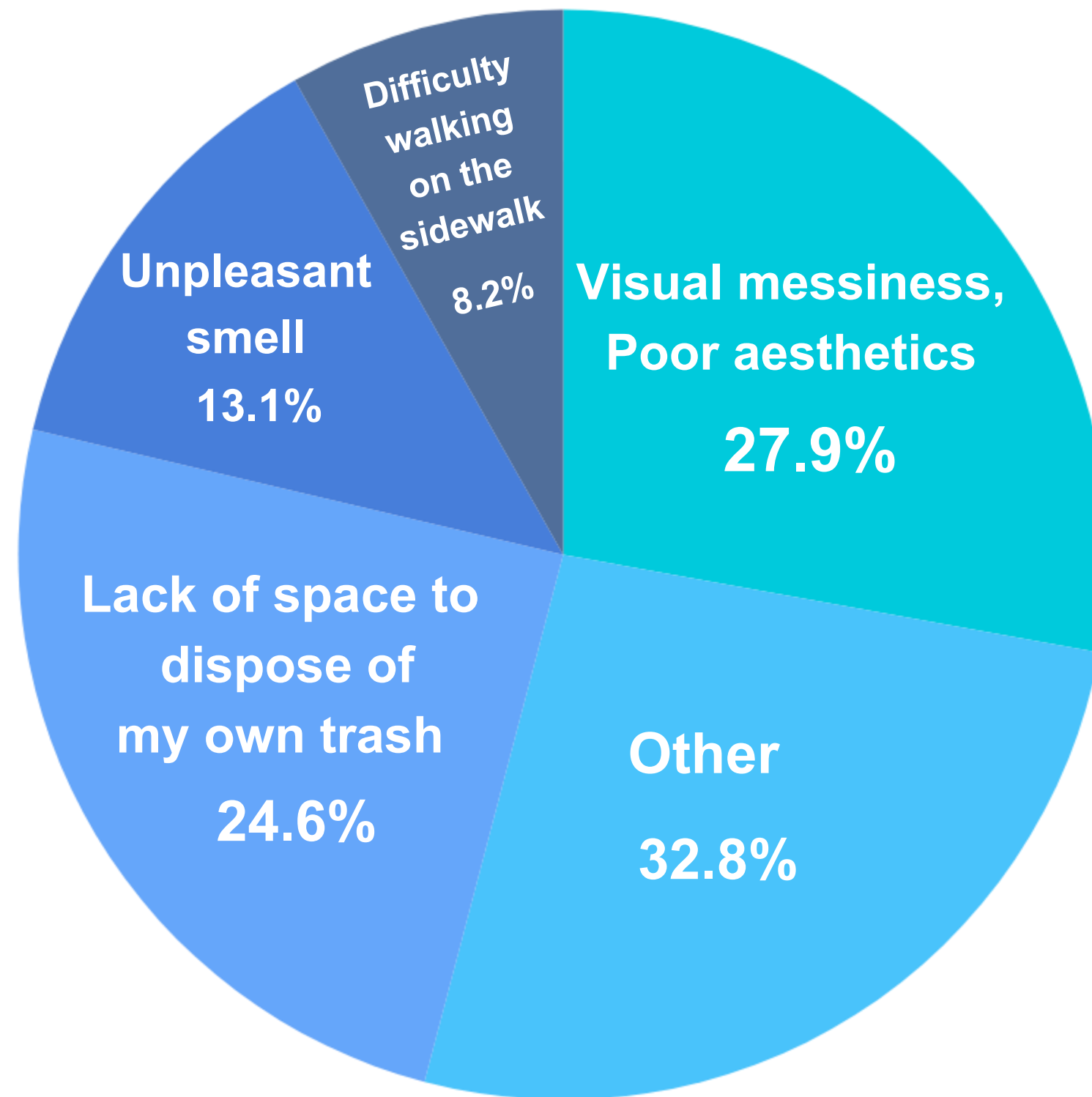
Hypothesis [1]



Do you feel that overflowing trash bins are a significant problem in the area?



What was the biggest issue?



1

People living in Seattle must be feeling inconvenienced
by the overflow of trash.



2 Using our product will improve waste collection efficiency.

Trash Bin Selection

Designate public trash bins and mark their locations on the map.

Recording load capacity with a 'human sensor'

Visit the same trash bin 2-3 times at regular intervals to visually record the load volume, noting the time and taking photos.

Full Collection vs. Selective Collection

Calculate and compare the distance/time for each method using Google Maps

HUMAN SENSOR



The reason for adopting this method

There are limitations to developing an actual system within the given timeframe.

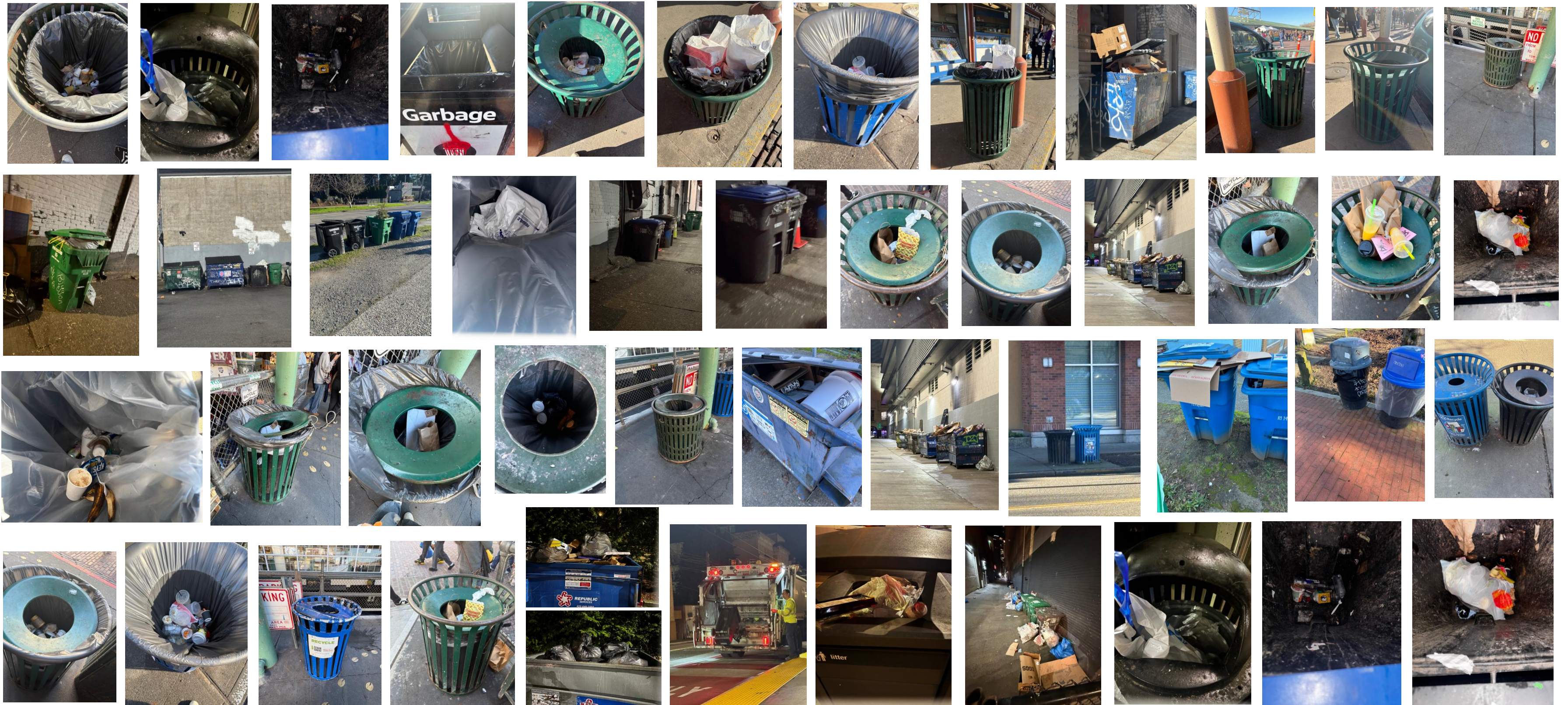
Even if the sensor hasn't been developed yet, to prove our hypothesis



“ **We acted as human sensors** ”

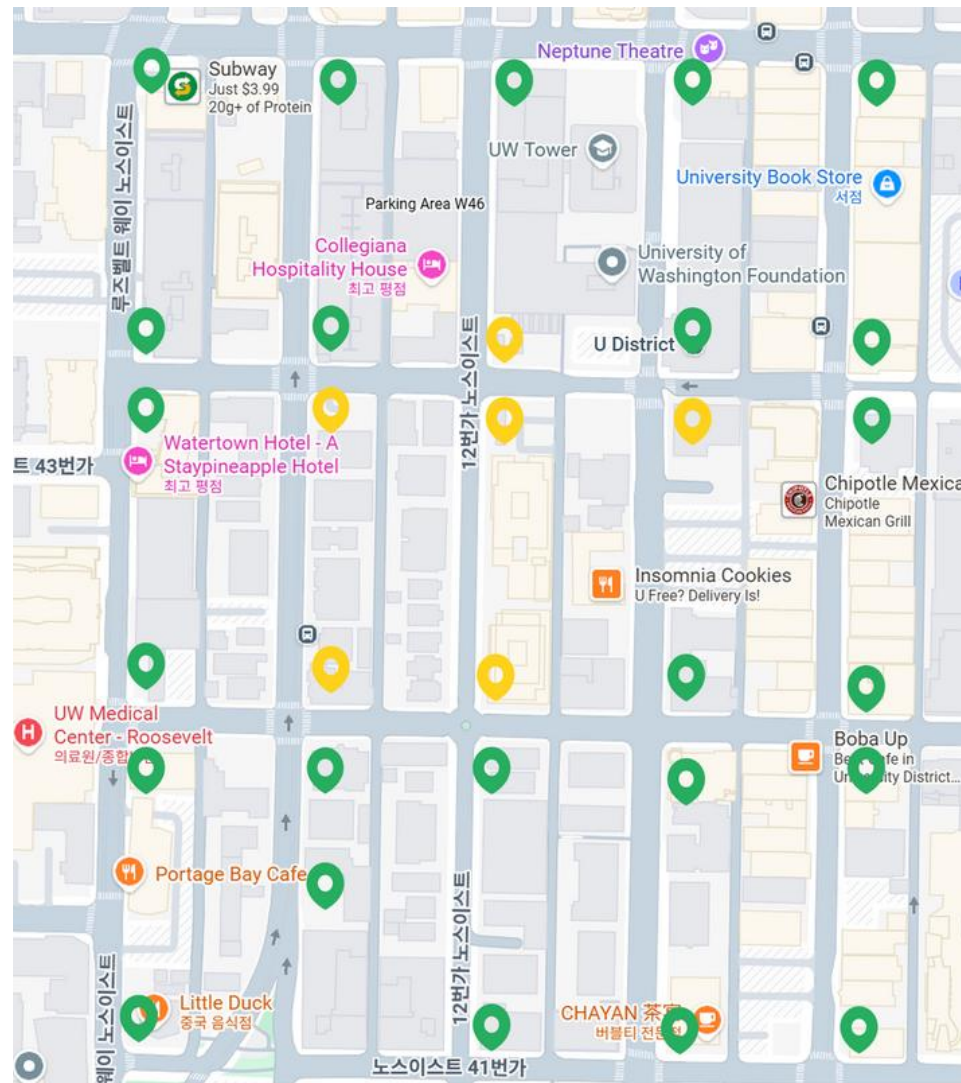
part 4

Hypothesis [2]

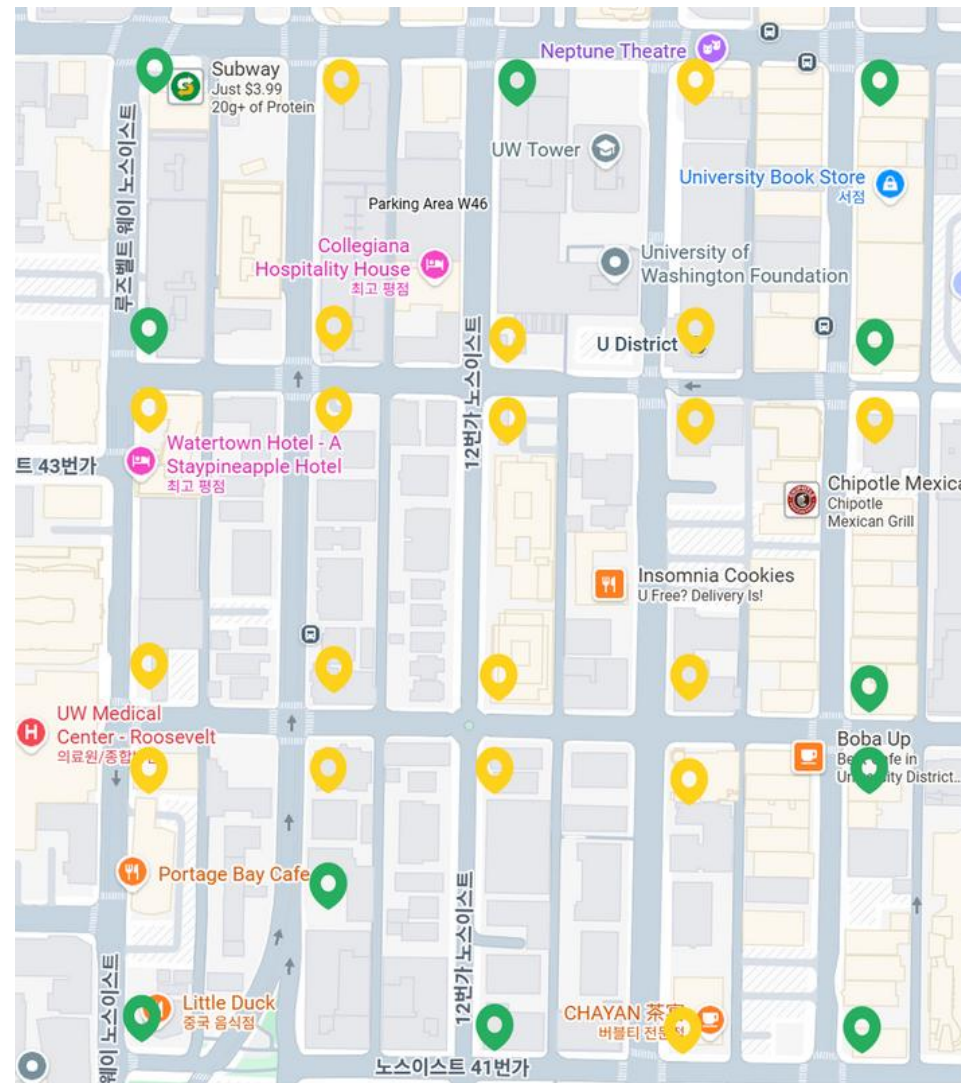
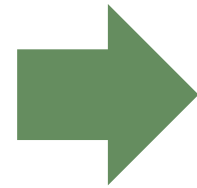


- High (60%~)
- Middle (30~59%)
- Low (~29%)

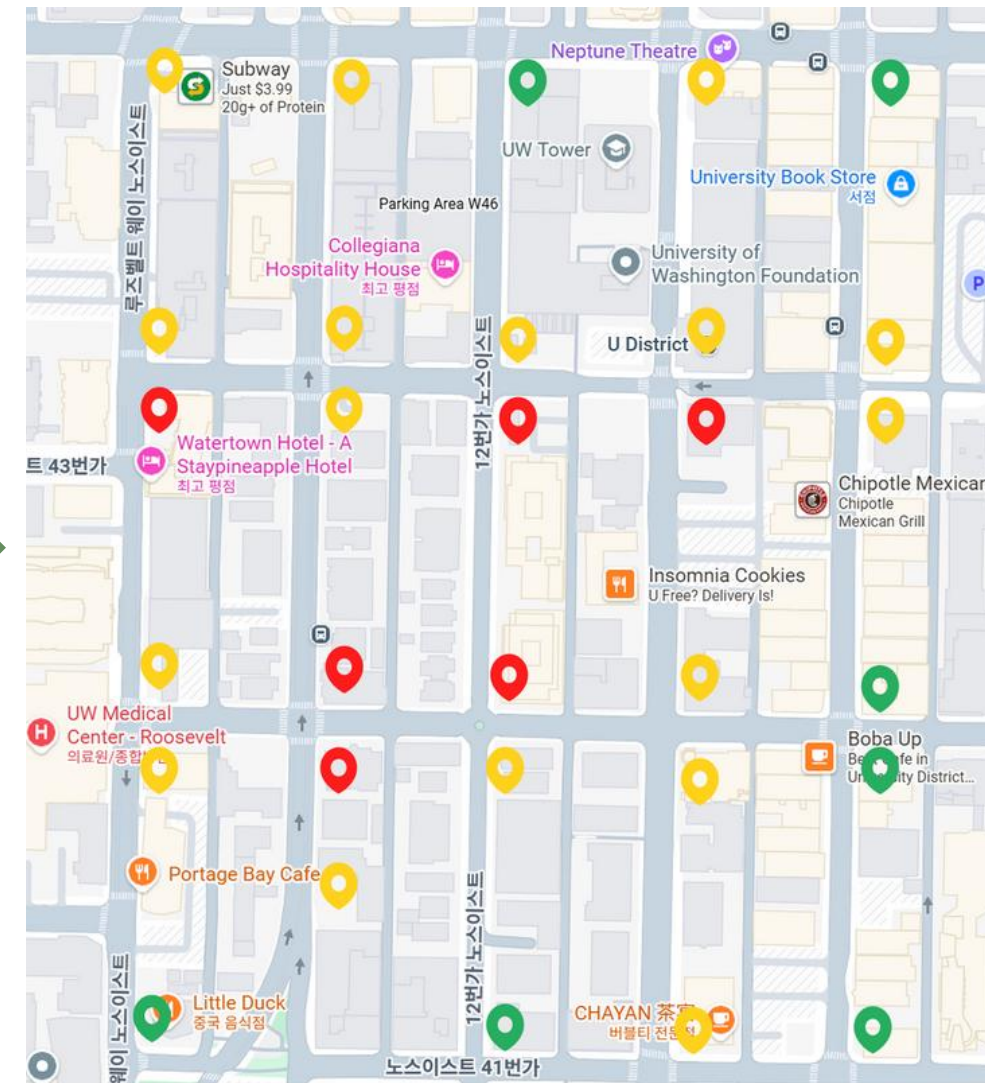
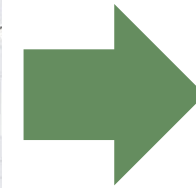
Hourly trash volume in the trash can



10:00

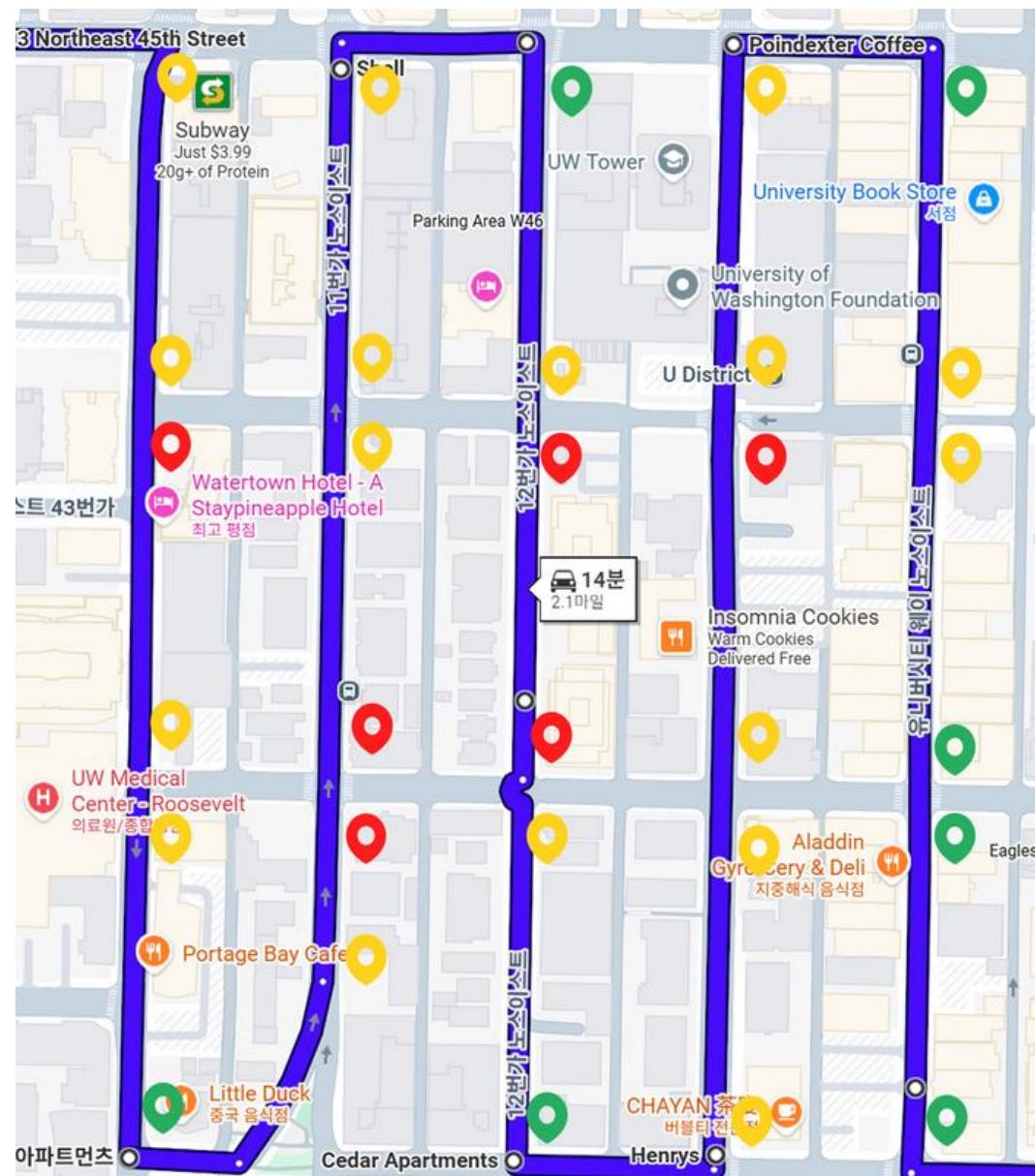


13:00



16:00

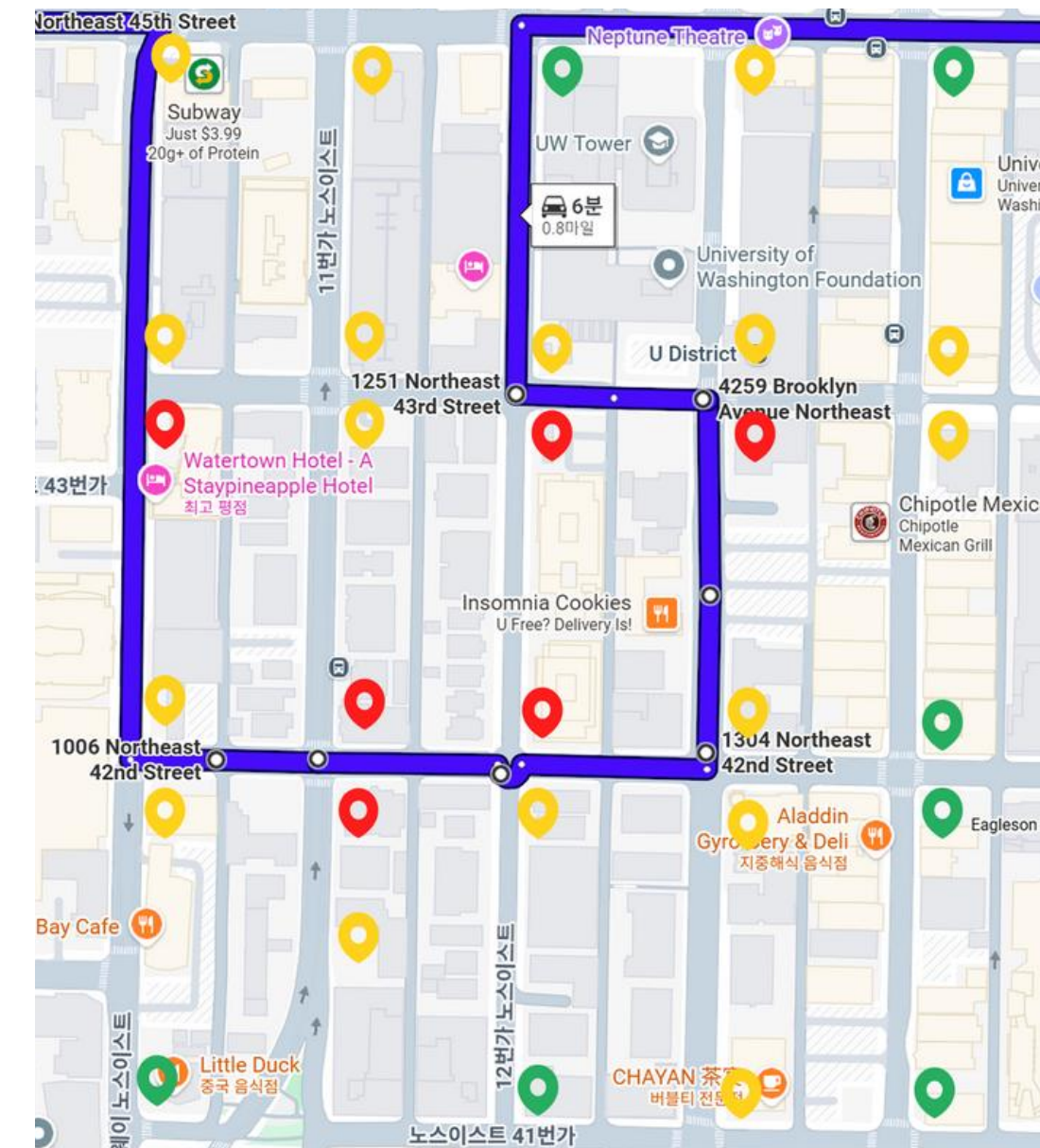
Full Collection



Travel Time: 14 min

Path
Optimization

Selective Collection



Travel Time: 6 min

2

Using our product will improve waste collection efficiency.



3

If our product can improve efficiency,
city governments and waste collection companies will use it.

Short User Reaction Test

Participants

- Campus facilities staff
- Cleaning staff



Scenario Shown

“Check bin fill levels
before collection”

Key Questions

- Unnecessary pickups?
- Would this improve efficiency?
- Would you try it as a pilot?

3

If our product can improve efficiency,
city governments and waste collection companies will use it.



Short User Reaction Test

Participants

- Campus facilities staff
- Cleaning staff



Scenario Shown

“Check bin fill levels
before collection”

Key Questions

- Unnecessary pickups?
- Would this improve efficiency?
- Would you try it as a pilot?

Before

Plan to verify language communication issues through on-site interviews in the U.S. (campus/hotel cleaning positions).

Issue

The school janitor was unavailable for an interview due to the weekend and Martin Luther King Jr. Day.



The hotel janitor was unavailable for an interview due to communication issues.



Consequently, we decided to give more weight to Hypothesis 3.

After

Due to on-site constraints, the interview location was shifted to within the country, and Hypothesis 3 was refined to focus on verifying API configuration/implementation.

Shifting Targets

Public holiday, access restrictions due to official
interview constraints This is really frustrating



Replaced with an interview with a Korean company

- 1 The respondent reported 1–2 instances in the past two weeks of unnecessary pickups (wasted trips), such as dispatching even when the bin was not sufficiently full.
- 2 They rated this issue as having a significant negative impact on costs and operations (score: 1).
- 3 They indicated they would be willing to use a tool that notifies only full bins and recommends visit priority/optimal routes and site access information, as it could reduce unnecessary pickups.

폐기물 관리자 관련 설문

1. 불필요한 픽업 경험 (빈도) 최근 2주간 통이 덜 찼거나 현장 진입 불가로 헛걸음한 적이 있나요?

(v)①있다. (☞ 1-1번으로)

②없다.

1-1. 헛걸음했던 이유가 무엇인가요?

설계 물량은 있었지만 실제 물량이 그만큼 나오지 않아, 최근 2주간 상차지에서 차량 1~2대 정도가 헛걸음으로 회차한 적이 있습니다.

2. 효율 개선 필요성 이런 헛걸음이 비용과 운영에 미치는 악영향을 1~5점으로 평가해 주세요.

매우 불만족(v)	다소 불만족	보통	다소 만족	매우 만족
✓ 1	2	3	4	5

3. 제품 사용 의향 수거 필요 현장만 골라 최적 경로와 진입 정보를 주는 도구가 있다면 쓰시겠습니까? (1~5점)

(v)①있다. (☞ 3-1번으로)

②없다. (☞ 3-1번으로)

3-1. 이유는 무엇입니까?

수거가 필요한 현장만 선별해서 보여주고, 최적 경로와 진입 정보까지 함께 제공된다면 불필요한 회차나 헛걸음을 줄일 수 있어 현장 운영에 실질적인 도움이 될 것 같아 사용 의향이 있습니다.

Limitation

1. Unvalidated sensor hardware

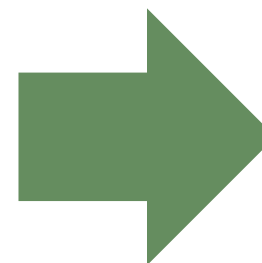
Need further testing to confirm sensor functionality

2. Limited long-term data

Insufficient data for robust trend analysis

3. Lack of cooperation from the city hall

No official city partnership yet, limiting on-site validation and adoption confirmation.



Next Step

1. Pilot testing of sensors

Conduct field tests to validate sensor performance

2. Accumulate long-term data

Gather sufficient data to analyze usage patterns

3. City Hall Cooperation Verification and Feasibility Assessment

Based on pilot results, we plan to collaborate with the city to assess on-site feasibility and adoption intent.

Dashboard Prototype





Q&A

Q&A